

Therefore the new gradient is 4 times that of the original graph.

14 Ans: D

Gases behave ideally when pressure is low.

Hence, consider the pV values when $p = 0$.

From the graph, when $p = 0$, $(pV)_{\text{boiling liquid}} = 5.95 \text{ kJ}$ and $(pV)_{\text{triple-point}} = 2.275 \text{ kJ}$

Using the equation, $\frac{T_{\text{boiling liquid}}}{T_{\text{triple point}}} = \frac{(pV)_{\text{boiling liquid}}}{(pV)_{\text{triple point}}}$

$$\frac{T_{\text{boiling liquid}}}{273.16} = \frac{5.95}{2.275}$$

$$T_{\text{boiling point}} = 714 \text{ K}$$

15 Ans: D